MeCP2 binding perturbs transcription of methylated DNA independent of chromatin

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Lay Abstract

DNA methylation adds chemical flags to some bases in the DNA strand. In human cells methylation turns genes off by stopping them making RNA copies from DNA sequences. MECP2 is a gene that codes for methyl-CpG binding protein 2 (MeCP2) and it helps methylation to do its job by binding to methylated DNA strands. Loss of MeCP2 causes a serious genetic brain disorder known as Rett syndrome. Scientists still do not understand, however, what MeCP2 does and why not having it leads to such a severe disease. To study this, we made a simple model in the bacteria *E. coli*. Using this we discovered that MeCP2 can affect the growth of *E. coli* on its own without any other factors. We found out that this is because MeCP2 interferes directly with the mechanisms that allow genes to be "expressed"- it causes some genes in *E. coli* to be turned on incorrectly, and some to be turned off. Our results could help us to understand why MeCP2 function is important for brain development and better understand the Rett syndrome disease.

Scientific Abstract

MECP2 is a gene encoding X-linked methyl-CpG binding protein 2 (MeCP2) and it binds to cytosine methylated at the 5 position (5mC). Loss-of-function of MeCP2 is a cause of Rett syndrome, a neurodevelopmental disorder. Loss of MeCP2 perturbs gene expression in the developing brain. However, the precise mechanism whereby MeCP2 affects transcription is unclear. In this study we tested the hypothesis that MeCP2 could interfere directly with transcription by binding methylated DNA and blocking transcription. To do this, we introduced MeCP2 into *E. coli*, either with or without 5mC methylation introduction by a highly active methyltransferase (mSSSI) which targets CG dinucleotides for methylation. Simultaneous 5mC and MeCP2 expression deteriorated the growth of *E. coli* by an average of 33%. Surprisingly we identified both up- and down- regulated genes genomewide. Gene Ontology results showed that the up- and down- regulated genes are associated with ribosome activity and energy metabolism respectively. Overall, differentially expressed genes (DEGs) had a higher CG proportion than non-differentially genes (non-DEGs). However within these DEGs, genes that are more strongly differentially expressed did not have a higher CG proportion. The length of genes did not have significant effect on the binding of MeCP2. Our results suggest that MeCP2 alone blocks RNA polymerase at some genes while promoting RNA polymerase activity at other genes. This may provide new insights into how loss of MeCP2 leads to Rett syndrome.

Background

In epigenetics, different sets of genes are turned on and off in different types of cells in the body without changing the underlying DNA sequence. 5mC of cytosine on the 5 position (5mC) is one form of epigenetic modification. MECP2 is a gene that provides instruction to make the protein MeCP2 known as methyl CpG binding protein 2 (Lewis et al., 1992). The MECP2 gene is found in the long (q) arm of

the X chromosome in band 28 ("Xq28"). MeCP2 protein is part of the methyl-CpG-binding domain protein (MBD) family but it possesses its own unique function which sets it apart from the group. It has two functional domains: the methyl-cytosine-binding domain (MBD) and the transcriptional repression domain (TRD) (Nan et al., 1993).

MeCP2 and Rett syndrome

Rett syndrome (RTT) is a rare genetic disorder that affects brain development, decreasing the capabilities in speech and motor functions which is then followed by cognitive and motor impairment, growth retardation and seizures (Caballero & Hendrich, 2005). It is mainly found in girls with a prevalence of about 1 in 10,000 (Fehr et al., 2011). RTT patients appear normal at birth to ~6 to 18 months as they could achieve the expected intellectual and motor milestones during this initial period (Hagberg et al., 1983; Percy & Lane, 2005). The symptoms will start to develop after this apparent normal development in which their language skills are lost and movement abnormalities are developed. RTT was initially considered as a female's disease due to its X dominance and lethality in hemizygous males (Zoghbi, 1988), but in rare cases some males can also be affected by Rett syndrome which is due to mutations in the MeCP2 (Meloni et al., 2000; Orrico et al., 2000).

MeCP2 is broadly expressed throughout the body, but it is most abundantly expressed in neurons. (Kishi & Macklis, 2004; Skene et al., 2010) Mutations in gene encoding MeCP2 leads to Rett syndrome (Chahrour & Zoghbi, 2007). During a period of postnatal brain development, the onset of symptoms in girls with RTT and mouse models with the disorder occurs in which the level of MeCP2 accumulates to exceedingly high levels in neurons of the brain, such that level of MeCP2 molecules in neurons approaches the level of nucleosomes in adult neuronal nuclei (Skene et al., 2010). Another research has shown that the level of this protein is low when the individual is at gestation stage, and it continues to increase its levels after the baby is born and up to the point where the child is 10 years old. This suggests that MeCP2 may become abundant once neurons have reached maturity (Shahbazian et al., 2002). RTT-like phenotypes in mice are highly associated with brain-specific disruption of MeCP2 which proved that RTT is predominantly a neuronal dysfunction disorder (Chen et al., 2001; Guy et al., 2001).

Function and Mechanism of Action

The MeCP2 protein plays a key role in regulating gene expression. Evidence suggests that it mainly acts as a transcriptional repressor (Nan et al., 1997). In the brain, expression of MeCP2 is highly associated with the maturation of central nervous system (CNS) and in maintaining and forming synapses between neurons (Luikenhuis et al., 2004). It is also believed that in additional to its role as a transcriptional repressor, MeCP2 is involved in alternative splicing in which it processes messenger RNA (mRNA) (Young et al., 2005). MeCP2 is important for brain function and development as loss of MeCP2 is a cause of Rett syndrome. Research has shown that MeCP2 function is required throughout

life for neurological health as deletion of MeCP2 in adult mice recapitulates the germlineknockout phenotype (McGraw et al., 2011). This suggests ongoing function of MeCP2 rather than just development, however the mechanism behind this is unclear.

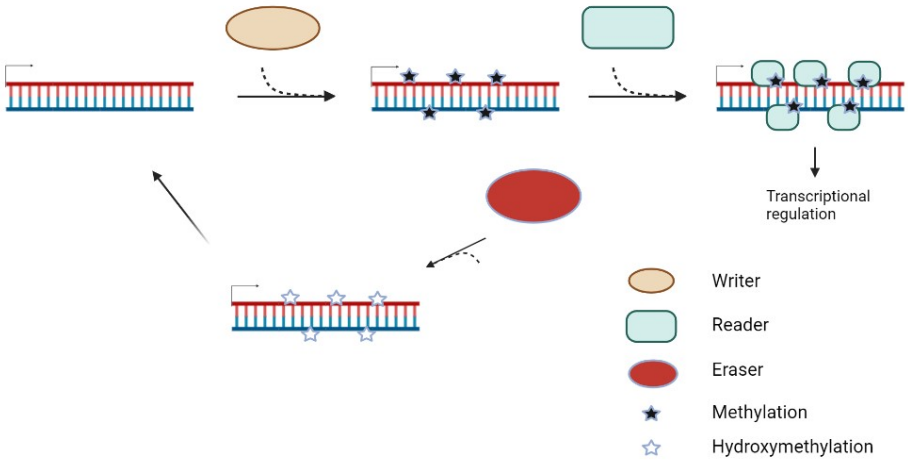


Figure 1. The process of DNA methylation (adapted from (Moore et al., 2013))

The mechanism of action of MeCP2 in humans follows the process of DNA methylation in which three classes of enzymes that add, recognise and remove 5mC are established: writers, readers and erasers (Moore et al., 2013). In humans the enzymes that introduce 5mC ("writers" in Figure 1) are DNMT1, DNMT3A and DNMT3B. DNMT3A and DNMT3B cause de novo methylation as they can introduce 5mC into unmodified DNA (Okano et al., 1999). DNMT1 then binds to "hemimethylated" DNA after replication of DNA containing 5mC. It then introduces 5mC into the unmethylated strand, enabling methylation to be propagated through cell division and thus acting as a "maintenance" methyltransferase (Law & Jacobsen, 2010). Readers are protein that can specifically bind to 5mC within DNA and then affect trasncription. In our case, the reader will be the MeCP2 protein. 5mC is removed by erasers which are typically TET enzymes. This is done by first modifying the DNA, creating a hydroxymethyl group before removing the methyl group (Rasmussen & Helin, 2016).

MeCP2 tends to bind to methylated CpG islands (mCpG) which are short-interspersed DNA sequences that are CG-rich (Lewis et al., 1992; Nan et al., 1997). They are typically 300-3000 base pairs in length and have been found in or near 40% of promoters of mammalian genes (Fatemi et al., 2005). In mammals, 70% to 80% of the CpG islands are methylated (Jabbari & Bernardi, 2004) and they are believed to control gene expression (Deaton & Bird, 2011). Once bound, it recruits corepressor complex containing SIN3A, NCOR and SMRT. This co-repressor complex then recruits histone deacetylase (HDAC) (Ehrhart et al., 2016; Gonzales et al., 2012) causing chromatin condensation around methylated DNA by removing acetyl groups from histone proteins (Szulwach et al., 2011). This then

causes methylated DNA to wrap tightly around histone proteins hence physically inhibiting the access of transcription factors hence resulting in gene silencing (Nan et al., 1997) However, the inhibition of histone deacetylase activity using drugs such as Trichostatin A only partially relieves transcriptional repression mediated by MeCP2 (Jones et al., 1998; Nan et al., 1998). This suggests an alternative pathway of repression by MeCP2 aside from recruitment of histone deacetylase. Most recent study has also found that MeCP2 may function as a transcriptional activator by recruiting the cAMP response element binding protein (CREB), a neuronal stimulus-dependent activator of gene transcription as majority of the genes appeared to be activated (Chahrour et al., 2008). This suggests the potential dual roles of MeCP2 in gene expression by acting as a key transcriptional regulator. However, it is unclear if these changes in genes are secondary in nature or regulated directly by MeCP2 (Cohen et al., 2008).

In MeCP2, the NCOR-SMRT interaction domain (NID) is located within the TRD domain and the interaction between MeCP2 and NCOR-SMRT is responsible for the transcriptional inhibition activity of MeCP2 (Lyst et al., 2013; Nan et al., 1998). Study showed that missense mutations in the methyl binding domain (MBD) of MeCP2 abolishes binding of MeCP2 to methylated DNA while mutations in the transcriptional repression domain (TRD) disrupts the binding of MeCP2 to NCOR/SMRT. This will affect the ability of MeCP2 to bridge the binding of MeCP2 to methylated DNA (Lyst et al., 2013). However, is it unknown why abolishing bridging between DNA and NCOR lead to Rett syndrome, one possibility might be lack of repression of downstream genes.

Despite being proven that MeCP2 functions as a silencer of transcription, identifying the specific binding targets of MeCP2 are difficult as MeCP2 binds broadly to the entire neuronal genome (Baker et al., 2013; Cohen et al., 2011; Gabel et al., 2015; Skene et al., 2010) and because changes in gene expression in the absence of MeCP2 are subtle (Baker et al., 2013; Chen et al., 2015; Gabel et al., 2015). The first large-scale mapping of MeCP2 binding sites in neurons has revealed that only 6% of the binding sites are CpG islands, 63% of MeCP2-bound promoters are actively expressed and that only 6% are highly methylated. This suggests that the primary function of MeCP2 is something other than silencing methylated promoters (Yasui et al., 2007).

In 2013, the lab of Joe Eckar and Margarita Behrens discovered that non-CpG methylation (mCH) is enriched in the postnatal brain of human and mouse model (Lister et al., 2013). Non-CpG methylation here refers to methylated cytosine that can be followed by any nucleotides – Adenine, Guanine or Thymine. They discovered that the percentage of methylated CH sites are elevated after birth and reaches maturity after almost 10 years and beyond. These observations remain interesting because the symptoms of Rett syndrome does not prevail until later on as the patients are born healthy. Until now it is known that DNMT3A is the sole “writer” of mCH in postnatal neurons (Gabel et al., 2015; Guo et al., 2014). One study has shown that the timing of when the mCH mark increases is very reminiscent to the

timing of the increase in MeCP2 levels. This suggests that MeCP2 might be the reader for this methyl mark (Chen et al., 2015). In addition, research has showed that long genes that contains high density of mCH are upregulated when MeCP2 and DNMT3A are disrupted as it showed a trend where longer genes have a greater extent of up-regulation in terms of its expression (Chen et al., 2015; Sugino et al., 2014). This suggests that MeCP2 acts at least in part as a transcriptional repressor by functioning through brain-enriched mCH to temper the expression of long genes in the brain. The striking three-way proportionalities between 5mC levels, MeCP2 binding-sites occupancy and transcriptional inhibition has been uncovered as the DNA binding-site specificity of MeCP2 showed that mCH and mCG both constitute target sites (Kinde et al., 2016; Shah & Bird, 2017). Other than that, a recent study has showed that MeCP2 also has additional interactions with chromatin by interacting with nucleosomes or in particularly through recognition of H3K27me3 and this interaction between MeCP2 and histone is involved in gene expression regulation (Lee et al., 2020).

Objective

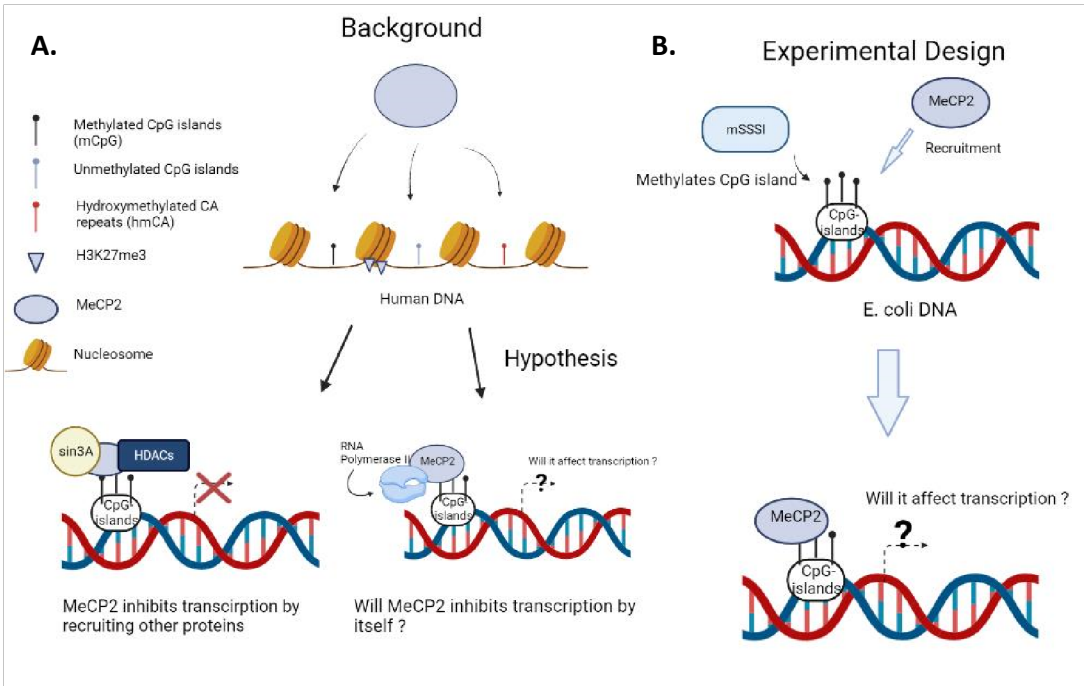


Figure 2. (A) The background of MeCP2. Left - MeCP2 represses transcription by binding to methylated CpG sites and recruit factors. Right – MeCP2 represses transcription by binding to methylated sites and collides with RNA polymerase II. (B) A figure of experimental design illustrating mSSSI acts as a “writer” by adding methyl groups to CpG sites while MeCP2 acts as a “reader” by binding to methylated CpG sites in *E. coli* DNA.

It is now thought that by recruiting HDAC proteins MeCP2 inhibits transcription and this could be through repressive chromatin set up by MeCP2. However, the molecular mechanism of the wild type protein and its mutant form remain elusive. Another simple alternative hypothesis is that the MeCP2 molecules themselves interferes with transcription by colliding with RNA polymerase II and this forms the basis of

our study. We suspect that MeCP2 alone is enough to affect transcription without other chromatin proteins (Figure 2A) and we tested this by in *E. coli* by observing if MeCP2 could affect the growth rate and gene expression of *E. coli* when combined with DNA methylation. The reason why *E. coli* is used is because chromatin is not found in *E. coli* so DNA is not bound to histone proteins in bacteria. In addition, co-repressor proteins are not found in bacteria so this could test if MeCP2 alone is sufficient to modulate gene expression. In this study, CpG methyltransferase (mSSSI) which is an active Dnmt which is used as a "writer" that adds methyl marks to CpG sites while MeCP2 will act as a "reader" that binds to methylated CpG sites and their synergy will be tested by looking at growth and gene expression of *E. coli* with and without 5mC. (Figure 2B). The truncated version of mSSSI (mSSSItc) which is a catalytically inactive version of mSSSI or known as an inactive Dnmt will be used as a control to make sure that the effect that we observed is due to methylation rather than the mSSSI protein itself without its catalytic activity.

Results

MeCP2 slows growth of *E. coli* in a manner dependent on MeCP2

We wanted to know whether MeCP2 could affect transcription by binding to methylated DNA in the absence of any interactions with chromatin. We therefore introduced MeCP2 into *E. coli*, which does not have histone proteins, by transforming with a plasmid containing the human MeCP2 DNA binding domain. Simultaneously, we introduced 5mC at CG sequences by expressing the prokaryotic methyltransferase mSSSI, which methylates CG dinucleotides in all sequence contexts (Renbaum et al., 1990). As a negative control, we also combined MeCP2 expression with the expression of a plasmid containing a catalytically inactive form of mSSSI (mSSSItc). Notably, MeCP2, mSSSI and mSSSItc were under control of IPTG, therefore we tested growth in the presence or absence of IPTG. We then tested whether growth was affected by each plasmid alone and in combination by measuring the growth continuously over 8 hours and then calculating the growth rate over that period for each condition.

We performed three biological replicates corresponding to separate colonies after transformation and repeated the entire experiment three times from separate replicate transformations. Plasmids showed a significant difference in growth in all experiments. ($p<0.0001$ for 3 experiments; Kruskal Wallis Test) [Figure 3A].

The effects of plasmids were most significant and comparable at 8th hour, demonstrating clear effects of IPTG induction. Following this we focused our analyses on bacteria induced with 0.5mM IPTG. mSSSI alone has marked effect on growth rate, whilst mSSSItc did not, indicating that CG methylation itself impairs growth in *E. coli*. MeCP2 only had a minor effect on growth, however, MeCP2 combined with mSSSI had a much stronger impact. Importantly, in all three experiments, the effects of mSSSI and

MeCP2 together was greater than the sum of the individual effects of MeCP2 and mSSSI, thus implying a synergistic effect (Figure 3B). The growth impairment of MeCP2 combined with mSSSI was significantly more than mSSSI alone (p-value: 5.03e-03, 3.36e-03, 3.24e02 for the 3 replicates; paired t-test with Bonferroni correction). Moreover, impairment of MeCP2 with mSSSI was significantly more than MeCP2 alone (p-value: 1.17e-08, 6.21e-06, 2.97e-04 from the three replicates; paired t-test with Bonferroni correction) and mSSSItc with MeCP2 (p-value: 5.24e-03, 3.50e-05, 1.35e-03 from the three replicates, paired t-test with Bonferroni correction). Taken together, these results demonstrate that MeCP2 could affect growth when CG methylation is introduced. Since the growth effect is not seen when MeCP2 is expressed in conjunction with a catalytically inactive version of mSSSI, this demonstrates that the growth impairment is due to the known ability of MeCP2 to specifically bind 5mC in the CG context.

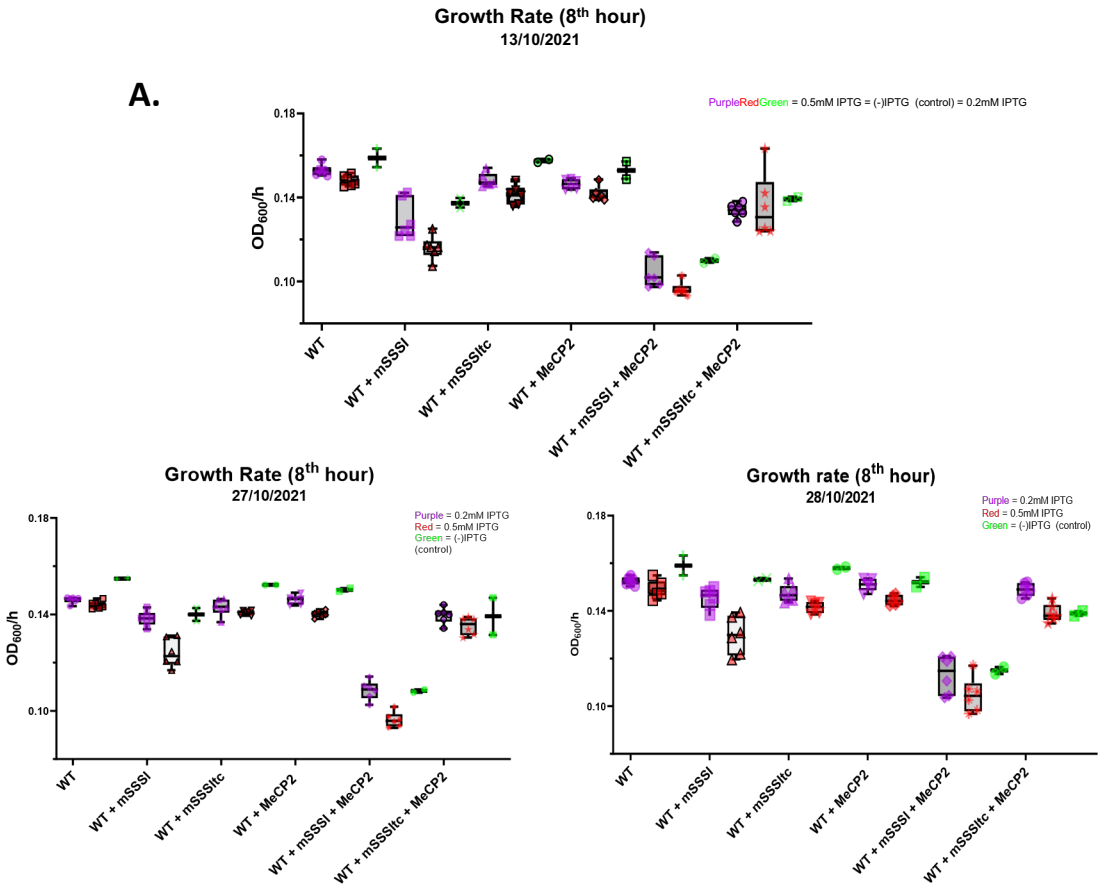
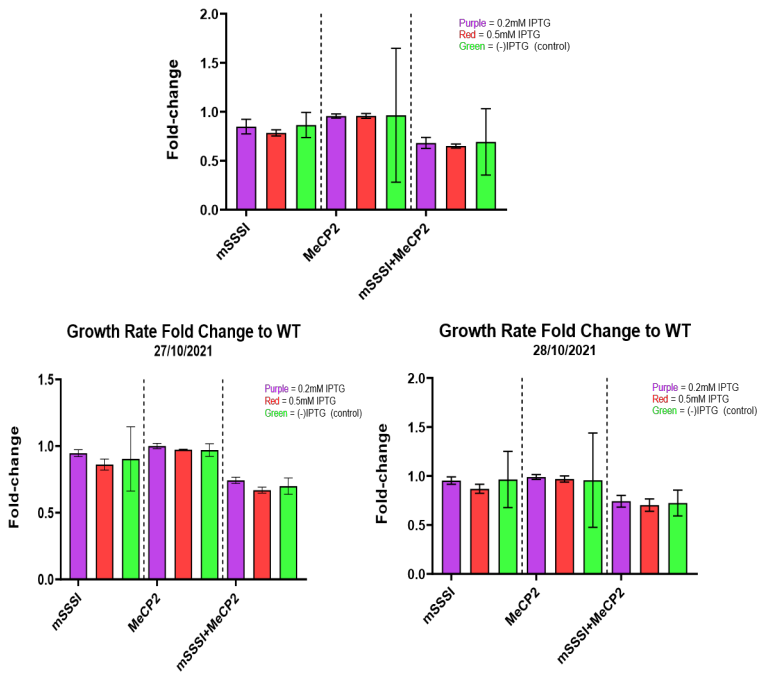


Figure 3. Growth curve experiments analysis. (A) Box and whiskers plot showing the growth rate of *E. coli* under different treatments at 8th hour across 3 biological replicates (left to right: Wild Type (WT), WT+mSSSI, WT+mSSSItc, WT+MeCP2, WT+mSSSI+MeCP2 and WT+mSSSItc+MeCP2). The line in the middle represents the median value, the box represents the 25th percentile and 75th percentile interquartile range (IQR) and the whiskers represent the maximum point no more than 1.5 times the IQR.



(B) The bar plots of average growth rate fold change of *E. coli* transformed with mSSSI, MeCP2 and mSSSI and MeCP2 relative to WT with 95% confidence interval error bars.

Following this, we further visualised the result of the growth curve experiments by constructing bar plots of the growth rate fold change of *E. coli* transformed with mSSSI, MeCP2 and mSSSI combined with MeCP2 relative to Wild Type *E. coli* (Figure 3B). Across the 3 biological replicates of growth curve experiments, the growth rate of *E. coli* transformed with mSSSI is decreased by an average of 16%, MeCP2 is slowed by 3% while mSSSI combined with MeCP2 is slowed by 33%. This suggests that the synergy between DNA methylation and MeCP2 did affect growth rate, however the mechanism remains unclear.

<u>MeCP2 and 5mC affects Gene Expression</u>

Next, we then wanted to know if MeCP2 is affecting growth by interfering with transcription without the need of histone proteins by studying if MeCP2 combined with mSSSI could affect the gene expression of *E. coli*. We investigated this by observing and analysing the RNA sequencing result of *E. coli* transformed with the prokaryotic methyltransferase mSSSI combined with MeCP2, the negative control mSSSItc combined with MeCP2 alongside with *E. coli* transformed with these plasmids alone.

Similar to the growth curve experiments, we did three biological replicates corresponding to separate colonies after transformation and repeated the entire experiment three times from separate replicates.

We used DESeq (Love et al., 2014) to identify differential expressed genes from the RNA-seq data by using the cutoff of log base 2-fold change (log2FC) of >1 for upregulated genes and <(-1) for downregulated genes, requiring an adjusted p-value (padj) of <0.01 in each condition. Principle component analysis (PCA) of the genes expressed in all the conditions showed a clear difference in gene expression by WT and mSSSI and MeCP2 compared to the other conditions. From the first two major axes of transcriptome variation, we found that genes affected by mSSSI and MeCP2 formed a cluster that has the greatest distance compared to WT especially on PC1 that accounts for 64% variance (Figure 3A), indicating that gene expression changes induced by mSSSI and MeCP2 are transcriptionally distinct from Wild Type expressed genes. Notably, the cluster formed by genes expressed in WT and mSSSI and MeCP2 deviates approximately twice the distance from Wild Type clusters when compared to the clusters formed by genes expressed in WT and mSSSI alone (Figure 4A).

Moving on, we observed that mSSSI alone did affect gene expression in cells as a total 341 genes were differentially expressed, whilst mSSSItc or MeCP2 alone only had a minor effect on gene expression as only a small number of genes are differentially expressed. However, cells expressing mSSSI combined with MeCP2 lead to a more profound effect as 686 genes were differentially expressed. Interestingly a similar number of genes were upregulated as downregulated (383 compared with 303). Cells expressing mSSSItc and MeCP2 lead to 55 differentially expressed genes (Figure 4B).

We further visualized the differentially expressed genes in different conditions by creating volcano plots samples at a log2FC cut-off of 1 or (-1), requiring a p-value of <0.01. The genes are highlighted in red if they pass both the log2FC cut-off and p-value threshold. We observed the most red-dotted genes that are differentially expressed by WT and mSSSI and MeCP2, followed by a modest number of red-dotted genes in cells expressing mSSSI. Cells expressing mSSSItc and MeCP2 alone had little to no effect on its gene expression. In the volcano plot of genes expressed in WT and mSSSI and MeCP2, lacZ is the top-ranked gene with a log2FC of 7.96 (Figure 3C).

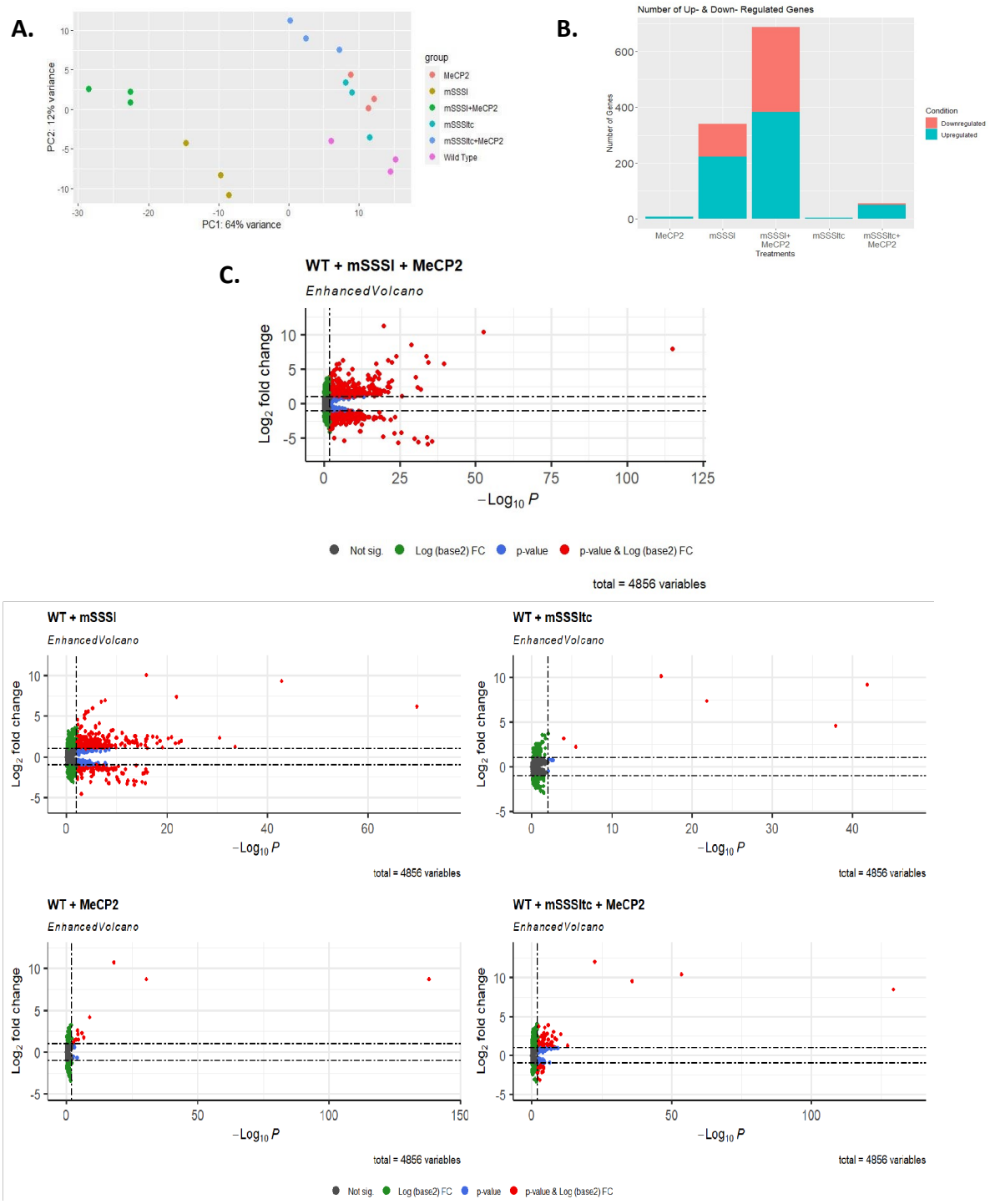


Figure 4. RNA-sequencing data analysis. (A) PCA plot with PC1 (64% variance) and PC2 (12% variance). (B) Number of up- and down- regulated genes in each condition (From left to right: MeCP2, mSSSI, mSSSI+MeCP2, mSSSItc, mSSSItc+MeCP2; blue – upregulated genes, red- downregulated genes) (C) Volcano plots of differentially expressed genes in different conditions, depicted with a cutoff of log base 2-fold change (log2FC) of >1 or <(-1) and adjusted p-value (p.adjusted) of <0.01. Red circles showed genes which have significantly differentially expressed. Blue circles showed genes that are significant, green circles showed genes that are differentially expressed and grey

circles showed genes that are not significantly expressed. -Log10 of p-value here is used for a better scaling and visualization.

We then wanted to identify genes that are affected specifically due to the synergy between MeCP2 combined with mSSSI by eliminating genes that were both differentially expressed in cells expressing mSSSI and MeCP2 and other plasmids. This set of non-overlapping genes, which we refer to as "affected genes in mSSSI and MeCP2 (non-overlaps)" consists of 169 upregulated genes and 197 downregulated genes. The top 20 most significantly differentially expressed genes are visualised in a hierarchical clustering heatmap. This analysis confirmed that these genes are mostly downregulated or no change in other conditions (Figure 5). The top ranked gene is *tdh* with a log2FC of -1.90. TDH encodes L-threonine 3-dehydrogenase (TDH) enzyme, which is rate limiting for the conversion of threonine into glycine and acetyl-CoA (Bowyer et al., 2009). We concluded that MeCP2 binding to methylated DNA has clear effects on *E. coli* gene expression.

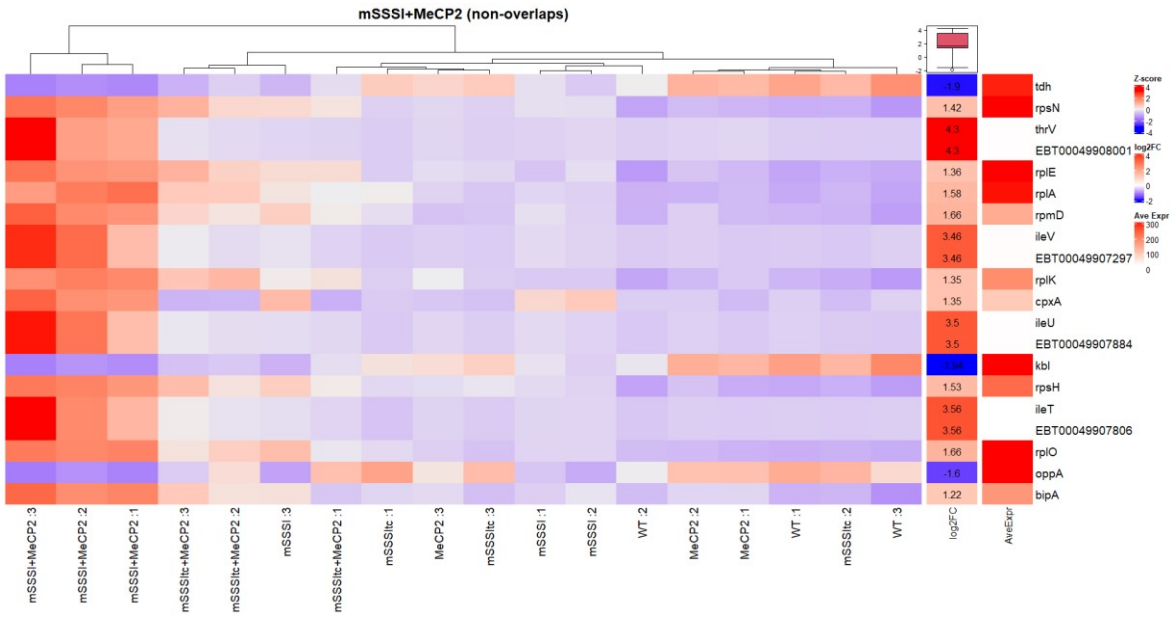


Figure 5. Hierarchical clustering heatmap of the top 20 differentially expressed genes in WT and mSSSI and MeCP2 (non-overlaps). Each colored cell on the map corresponds to a concentration value (Red- upregulated, Blue – downregulated, White – no change).

<u>Upregulated Genes are Associated to Ribosome Activity while Downregulated Genes are Related to Energy Metabolism</u>

Having demonstrated that MeCP2 binding to methylated DNA affected gene expression, we next wanted to investigate whether this might explain the effect of MeCP2 on growth rate. We therefore performed Gene Ontology overrepresentation analysis on genes specifically affected by MeCP2 and

mSSSI. Of the 169 upregulated genes, biological process analysis showed that genes that were significantly enriched (p-value <0.05) were related to ribosome and ribonucleoprotein complex biogenesis, organelle organization, translation, peptide biosynthetic process and the assembly of different organelles and complexes. Subcellular component analysis showed significant enrichment in genes involved in intracellular non-membrane bound organelle, intracellular organelle, non-membrane bound organelle, organelle, ribonucleoprotein complex, ribosome and different types of ribosomal subunits. Molecular function showed enrichment for ribosome activity, RNA binding and serine activity (Figure 6).

Overrepresentation analysis of the 197 downregulated genes showed that these genes were primarily involved in energy metabolism. Biological process analysis showed that these genes are involved in generation of precursor metabolites and energy, energy derivation by oxidation of organic compounds, respiration, oxidative phosphorylation, metabolic processes and threonine processes. Subcellular component analysis showed significant enrichment in genes involved in energy metabolism complexes such as plasma membrane protein complex, membrane protein complex, catalytic complex, respiratory chain complex etc. These genes were mostly important for oxidoreducatase activity, electron transfer activity, transmembrane transporter activity and the activity of NADH and NADPH. (Figure 6). Reduced expression of genes involved in energy metabolism might explain the reduced growth of *E.coli* when MeCP2 is expressed in conjunction with methylation of DNA.

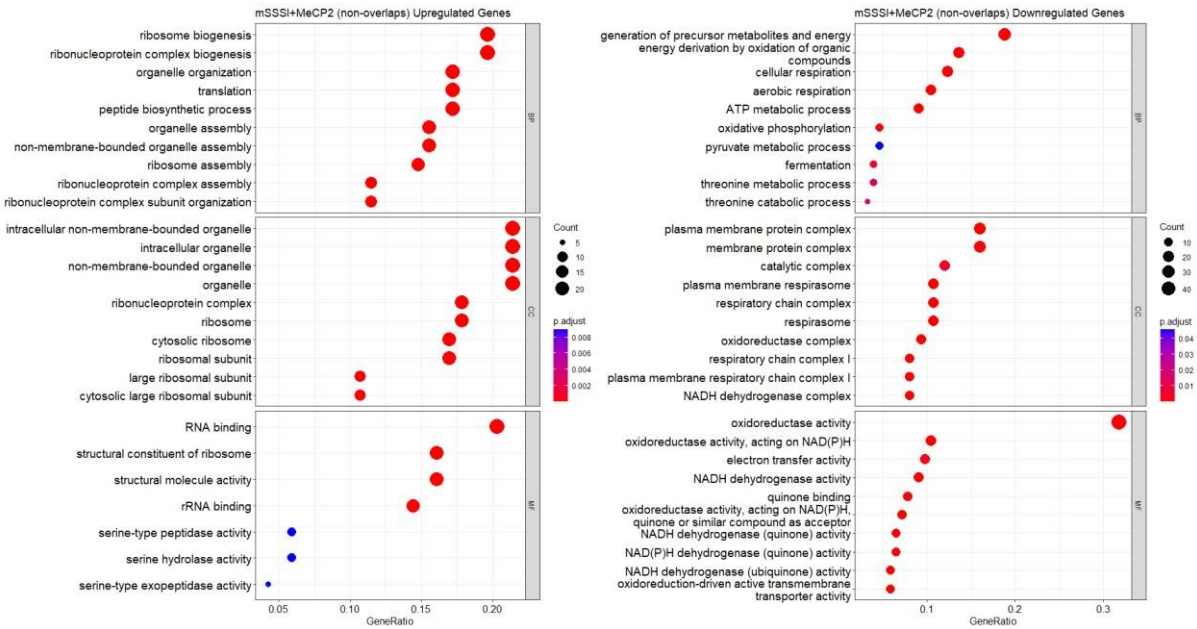


Figure 6. The top 10 ranked GO terms of upregulated genes (left) and downregulated genes (right) in cells expressing mSSSI and MeCP2 according to gene count. “P-adjust” is the p-value adjusted using the Benjamini-Hochberg procedure. ‘Gene count’ is the number of genes enriched in a GO term. The

X axis represents the gene ratio which is the percentage of total DEGs in the given GO term whereas the Y axis represents the Gene Ontology functional classification. GO – gene ontology, BP - biological processes, CC – cellular component and MF – molecular function.

Differentially Expressed Genes have Higher CG content than non-Differentially Expressed Genes (non-DEGs)

MeCP2 is known to bind to 5mC while mSSSI methylates cytosine in CG sequence contexts (Lewis et al., 1992; Nan et al., 1997). Therefore, we tested whether genes that were affected by expressing MeCP2 and mSSSI together had an increased proportion of CG sequences, thus more likely to have 5mC that could be bound by MeCP2. We studied this by constructing boxplots of the CG proportion of affected genes in cells expressing mSSSI, mSSSI and MeCP2, the negative control mSSSItc and MeCP2 as well as in mSSSI and MeCP2 (non-overlaps) and comparing them to the CG proportion of non-differentially expressed genes (non-DEGs). We observed that the upregulated genes in cells expressing mSSSI alone and mSSSI combined with MeCP2 have significantly higher CG proportion than non-DEGs (p-value: 3.30e-05, 3.30e-09; Wilcoxon unpaired test with Bonferroni correction). Genes specifically affected in mSSSI and MeCP2 also have significantly higher CG proportion than non-DEGs (p-value: 2.44e-03; Wilcoxon unpaired test with Bonferroni correction), whilst the negative control mSSSItc and MeCP2 do not (p-value: 1.00; Wilcoxon unpaired test with Bonferroni correction) [Figure 7A]. This suggests that the genes affected have an increased proportion of CG sequences consistent with the hypothesis that MeCP2 binds to these genes.

We next compared the CG proportions of downregulated genes to non-differentially expressed genes (non-DEGs). The genes affected in the negative control mSSSItc combined with mSSSI were not compared as the number of genes affected were not sufficient to provide useful analysis. There was no significant difference in CG proportion of downregulated genes in cells expressing mSSSI alone as well as mSSSI combined with MeCP2 to non-DEGs. (p-value: 1.00, 5.97e-01; Wilcoxon unpaired test with Bonferroni correction) The genes affected specifically in mSSSI and MeCP2 also have no significant difference in CG proportion when compared to non-DEGs (p-value: 1.00; Wilcoxon unpaired test with Bonferroni correction) [Figure 7A]. This indicates the genes downregulated in terms of expression did not have increased proportion of CG sequences.

CG content is Not Correlated to the extent of Gene Expression Change

Having established that genes upregulated when MeCP2 and mSSSI were expressed had a higher CG proportion, we wanted to see if CG content itself was correlated to gene expression change, in other words whether genes with higher CG content amongst upregulated or downregulated genes were likely to show greater changes in expression than those with lower CG content. We studied this by focusing on the genes affected specifically by mSSSI combined with MeCP2. The log2FC of the affected genes

were divided equally into 5 bins and genes were allocated into different bins based on their log2FC and the CG proportion of each bin is compared to non-DEGs.

For upregulated genes in cells expressing mSSSI and MeCP2, we found that only genes in bin 3 where their log2FC falls in the range of 40-60% were significantly higher in CG proportion than nonDEGs. (p-value: 0.026; Wilcoxon unpaired test with Bonferroni correction). We then performed Gene Ontology overrepresentation analysis with the affected genes in each bin and we observed that genes in bin 3 returned GO terms that were similar to the top 10 ranked GO terms returned from performing Gene Ontology analysis using the entire set of genes specifically upregulated in cells expressing mSSSI and MeCP2 (Figure 7B), suggesting this subset of genes were associated with GO terms that are significantly enriched.

Meanwhile for downregulated genes in cells expressing mSSSI and MeCP2, we observed that surprisingly, genes affected in bin 5 where their log2FC falls in the highest range of 80-100% were significantly lower in CG proportion than non-DEGs (p-value: 0.023, Wilcoxon unpaired test with Bonferroni correction). We observed that affected genes in bin 1 and bin 2 were associated with energy metabolism as the genes in bin 1 and bin 2 returned GO terms that mostly resembles the top 10 ranked GO terms returned using all downregulated genes from mSSSI and MeCP2 (non-overlaps) [Figure 7C], suggesting only genes in bin 1 and bin 2 were likely to be associated with GO terms that are significantly enriched.

We also observed that there was no significant difference in the CG proportion of genes in bin 1 compared to genes in bin 5 for both up- and down- regulated genes in cells expressing mSSSI and MeCP2 (p-value: 0.16, 0.69 for up- and down- regulated genes; Wilcox test) [Figure 7B; Figure 7C]. This suggests that CpG content most likely does not determine if MeCP2 and DNA methylation will result in a high or modest regulation of these genes.

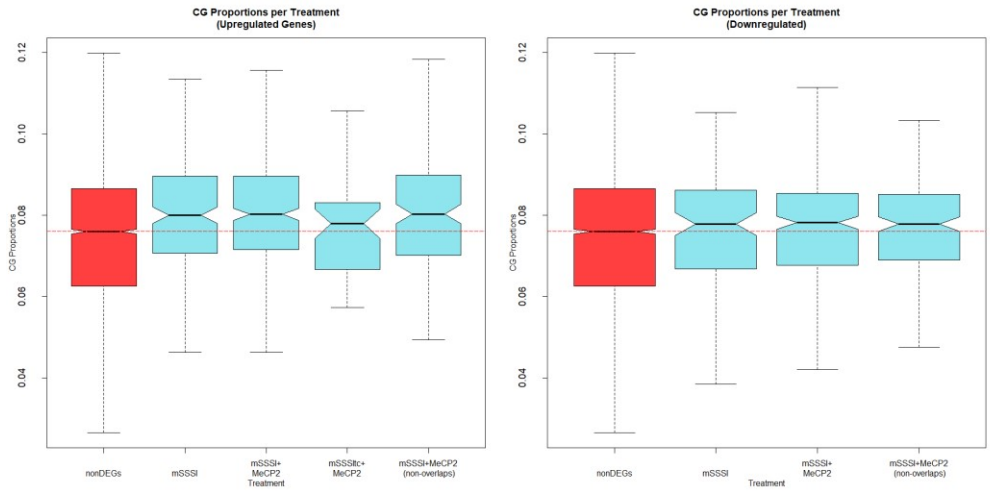


Figure 7. CG analysis. (A) Box and whiskers plot of CG proportion of upregulated genes (left) and downregulated genes (right) in cells expressing mSSSI, mSSSI and MeCP2 and mSSSItc and MeCP2 compared to non-differentially expressed genes (nonDEGs). mSSSI and MeCP2 (nonoverlaps) refers to genes affected specifically by mSSSI and MeCP2. The red dotted line represents

the median CG proportion of nonDEGs, the line at middle represents median, the box represents 25th and 75th percentile interquartile range (IQR), the whisker is the maximum point no more than 1.5 times the IQR and the notch represents 95% confidence interval on the median.

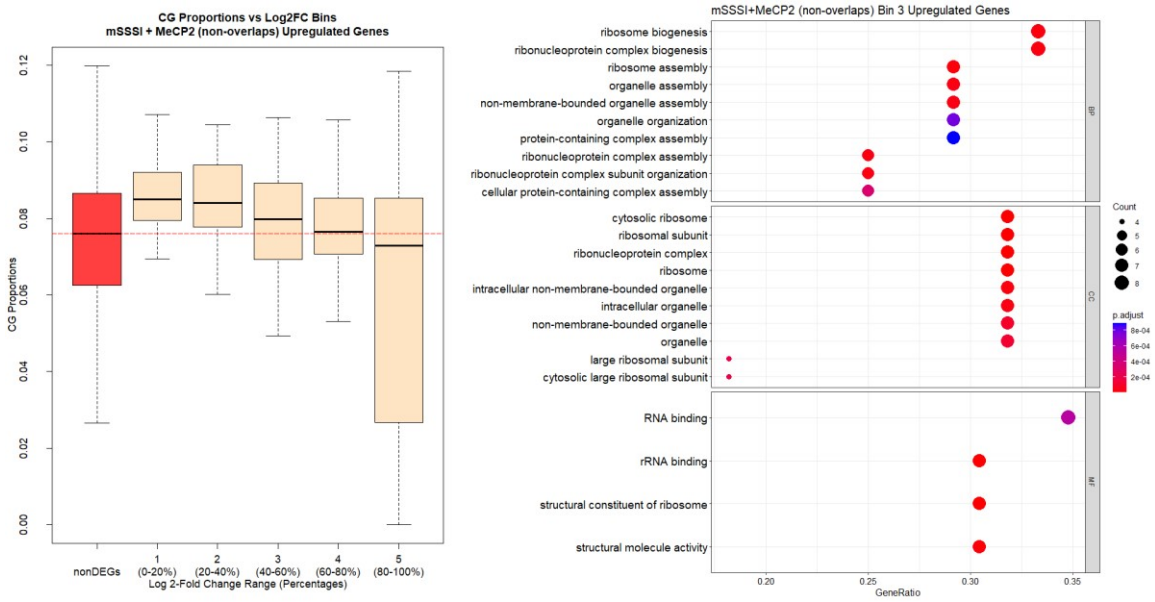


Figure 7. (B) Box and whiskers plot of CG proportion of upregulated genes affected in WT and mSSSI and MeCP2 (non-overlaps) according to different range of log2FC divided equally into 5 bins (1 – lowest; 5 – highest) and compared to the CG proportion of nonDEGs (left). The top 10 ranked GO terms of genes in bin 3 according to gene count with p-value cutoff of <0.05 (adjusted using the

Benjamini-Hochberg procedure) (right). The red dotted line represents the median CG proportion of nonDEGs, the line at middle represents median, the box represents 25th and 75th percentile interquartile range (IQR) and the whisker is the maximum point no more than 1.5 times the IQR.

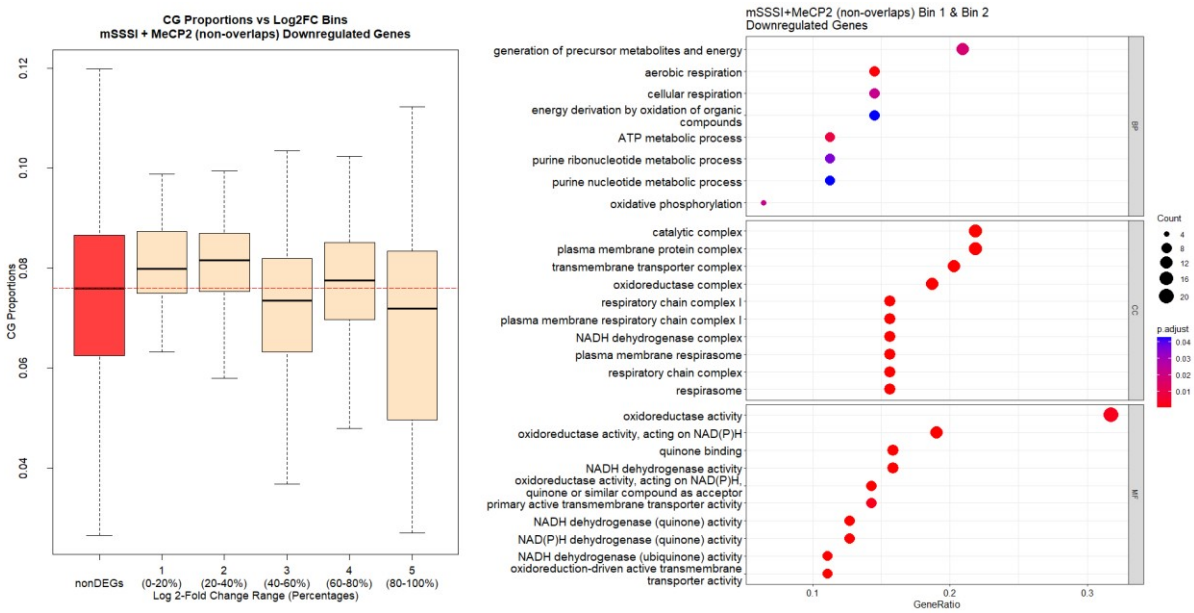


Figure 7. (C) Box and whiskers plot of CG proportion of downregulated genes affected in WT and mSSSI and MeCP2 (non-overlaps) according to different range of log2FC divided equally into 5 bins (1 – lowest; 5 – highest) and compared to the CG proportion of nonDEGs (left). The top 10 ranked GO terms of genes in bin 1 and bin 2 according to gene count with p-value cutoff of <0.05 (adjusted using the Benjamini-Hochberg procedure) (right). The red dotted line represents the median CG proportion of nonDEGs, the line at middle represents median, the box represents 25th and 75th percentile interquartile range (IQR) and the whisker is the maximum point no more than 1.5 times the IQR.

<u>WT and mSSSI and MeCP2 has Higher Proportion of CG stretches than non-Differentially Expressed Genes (non-DEGs)</u>

MeCP2 is known to bind to CpG islands, which are long tracts of CG sequences (Nan et al., 1997). We therefore wondered whether genes affected by MeCP2 and mSSSI in *E. coli* would be more likely to have long stretches of CG. We studied this by comparing the proportion of CG stretches with different number of continuous CG counts in genes affected specifically by mSSSI combined with MeCP2 to non-DEGs. We found that stretches fewer than 8CGs in length were found in almost all genes whilst >18CGs stretches were found in too few genes for meaningful analysis [Figure 8A], so we decided to compare CG stretches with 8, 10, 12, 15 and 17 continuous CG sequences.

We observed that upregulated genes affected specifically by mSSSI and MeCP2 have significantly higher proportion of CG stretches with 8, 10 and 12 continuous CG than non-DEGs (p-value: 0.016, 0.026, 0.014, Wilcox test with Bonferroni correction). On the other side, downregulated genes did not have significant difference in proportion of CG stretches with 8, 10 and 12 continuous CG when compared to non-DEGs (p-value: 0.23, 1.00, 1.00, Wilcox test with Bonferroni correction) [Figure 8B].

This suggests that MeCP2 is likely to bind to CpG islands with 8 to 12 continuous CG sequences on a gene and upregulates their expression. Most genes that have CG stretches with 15 and 17 continuous CG were non-differentially expressed and they were classified as outliers in the boxplots and so there were not enough genes to produce meaningful analysis.

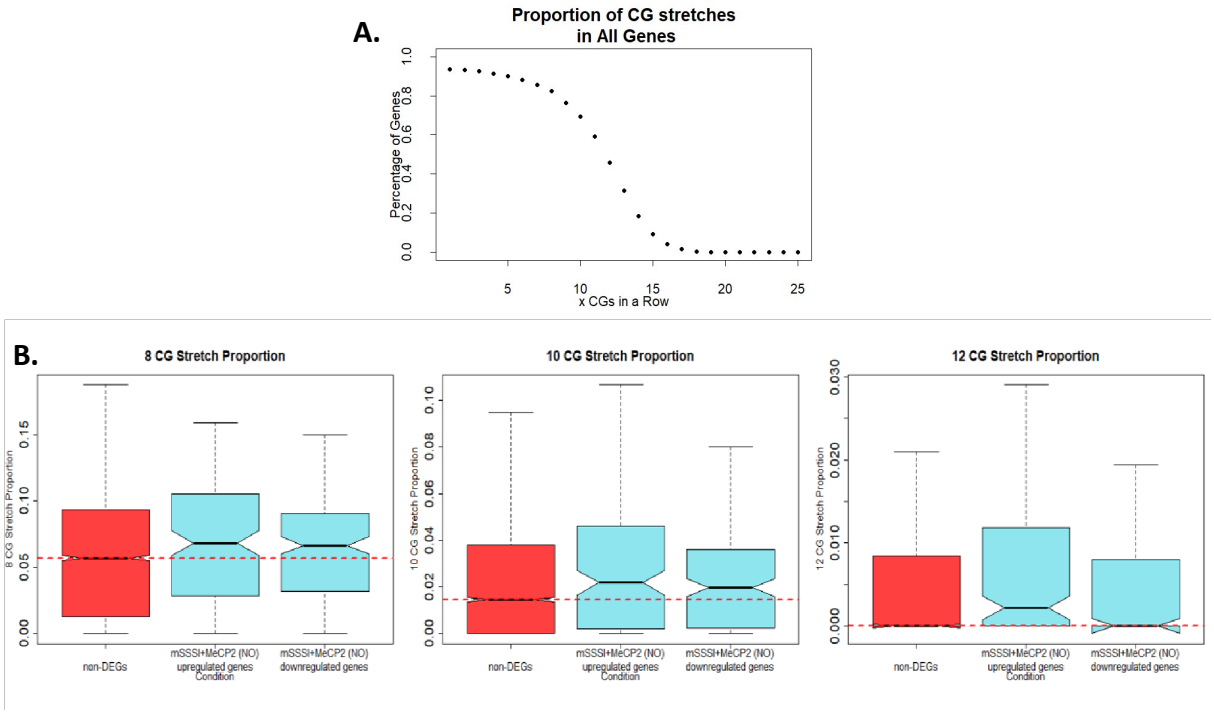


Figure 8. (A) The percentage of all genes (n=4856) with proportion of CG stretches with x CG sequences in a row (x = 1 to 25). (B) Box and whiskers plot showing the proportion of CG stretches with 8, 10 and 12 CG sequences in a row in different conditions (left to right: nonDEGs, mSSSI+MeCP2 (NO - non-overlaps; upregulated genes), mSSSI+MeCP2 (NO - non-overlaps; downregulated genes)). The red dotted line represents the median CG proportion of nonDEGs, the line at middle represents median, the box represents 25th and 75th percentile interquartile range (IQR), the whisker is the maximum point no more than 1.5 times the IQR and the notch represents 95% confidence interval on the median.

<u>Length of Genes Does Not Affect Binding of MeCP2</u>

MeCP2 has previously been suggested to regulate expression of long genes expression in mammalian brain (Gabel et al., 2015). We therefore tested whether long genes were more likely to be affected by MeCP2 by comparing the length of the affected genes to nonDEGs. We observed that only genes affected mSSSItc combined with MeCP2 have significantly lower gene length than nonDEGs (p-value:

9.92e-04; Wilcox test with Bonferroni correction). We further investigate if gene length correlates to gene expression by dividing the log2FC equally into 5 bins and classified genes into each bin based on their log2FC, which we then construct boxplots of the proportion of CG sequences for each bin and compare to non-DEGs. The results showed no significant difference in CG proportion in all bins for both up- and down- regulated genes to nonDEGs. This suggests MeCP2 does not bind and regulate specifically to long genes.

Discussion

In this study we set out to test the hypothesis that MeCP2 could affect gene regulation through binding to 5mC in DNA and affecting RNA polymerase activity directly, without any requirement for chromatin. We showed that MeCP2 expression in *E. coli*, alongside methylation of DNA in the CG sequence context, led to a reduction in growth, and was associated with gene expression changes. Below, I will discuss the extent to which these results support our initial hypothesis, and their implications for understanding the mechanism of MeCP2 activity in human cells.

1. The effect of MeCP2 on transcription

Our results demonstrated that MeCP2, expressed when CG sequences are methylated, is sufficient to retard growth in *E. coli*. This effect is likely due to the fact that MeCP2 expression leads to downregulation of genes involved in energy metabolism. Since MeCP2 alone does not have this effect, we speculate that this direct binding of MeCP2 to 5mC genome interferes with RNA polymerase directly. Supporting this, genes affected by MeCP2 were more likely to contain long CG stretches, so MeCP2 would be more likely to bind to them. However, it should be noted that many genes affected by MeCP2 especially downregulated genes did not have high proportions of CG sequences. Indeed, genes with the highest expression changes were depleted in CG dinucleotides relative to unaffected genes. This could be due to secondary effects whereby changes in expression of direct targets lead to further gene expression changes. One way to test this would be to use Global Run On- sequencing (GRO-seq), which measures transcription initiation (Core et al., 2008), following induction of MeCP2 expression as this would allow us to track the expression changes over time and distinguish primary and secondary effects.

Our model fails to explain why MeCP2 expression results in a similar number of up- and down- regulated genes. One possibility is that upregulated genes are secondary effects. However, the proportion of CG stretches in up- and down- regulated genes affected in WT and mSSSI and MeCP2 were similar, this does not seem to be a likely explanation. A further possibility is that the position of CGs relative to the start of the gene might be important (Figure 8). We speculate that if the CpG sites are close to the start of the gene then they block expression, but at the end of the gene they increase the expression instead

by accelerating the rate at which RNA polymerase terminates transcription. Another explanation could also be related to the organisation of *E. coli* as multiple genes are located on the same RNA. Perhaps the genes that are at the beginning of the operon are differentially expressed than those at the end of an operon. Further analysis of the characteristics of up- and down- regulated genes would be important to resolve this.

One important addition experiment that could be useful in explaining what causes MeCP2 expression to target particular genes would be to directly map where MeCP2 binds methylated DNA in *E. coli*. It could be, for example, that MeCP2 has specific sequence preferences beyond simply CG stretches and this might explain which genes are affected. ChIP-seq to map MeCP2 binding sites would be a useful experiment to perform (Johnson et al., 2007).

Implications for the function of MeCP2 in mammals

MeCP2 has been shown to have subtle effects on the gene expression in mammals, including modest downregulation of genes. (Kaludov & Wolffe, 2000) Mechanistically this has been attributed to the recruitment of chromatin modifying enzymes by MeCP2. However, our results suggest that MeCP2 can affect transcription alone, independent of chromatin structure. It will be interesting to test whether this contributes to MeCP2 activity in mammalian cells.

A key finding from mammals is that MeCP2 is more likely to affect the expression of long genes (Sugino et al., 2014). *E. coli*, however, we did not find any bias for MeCP2 to regulate long genes. The preference of MeCP2 to regulate long genes in mammals may therefore be due to chromatinrelated functions of this gene rather than direct effects on transcription.

Overall, our results provide some support for the notion that MeCP2 can affect transcription directly, potentially by interfering with RNA polymerase progression. Further work will be required to understand exactly why certain genes are more likely to be affected by direct MeCP2 binding. However, our results provide an important new angle to explore in understanding how MeCP2 affects transcription in mammalian cells and thus provide some new ideas about how loss of MeCP2 activity leads to Rett syndrome.

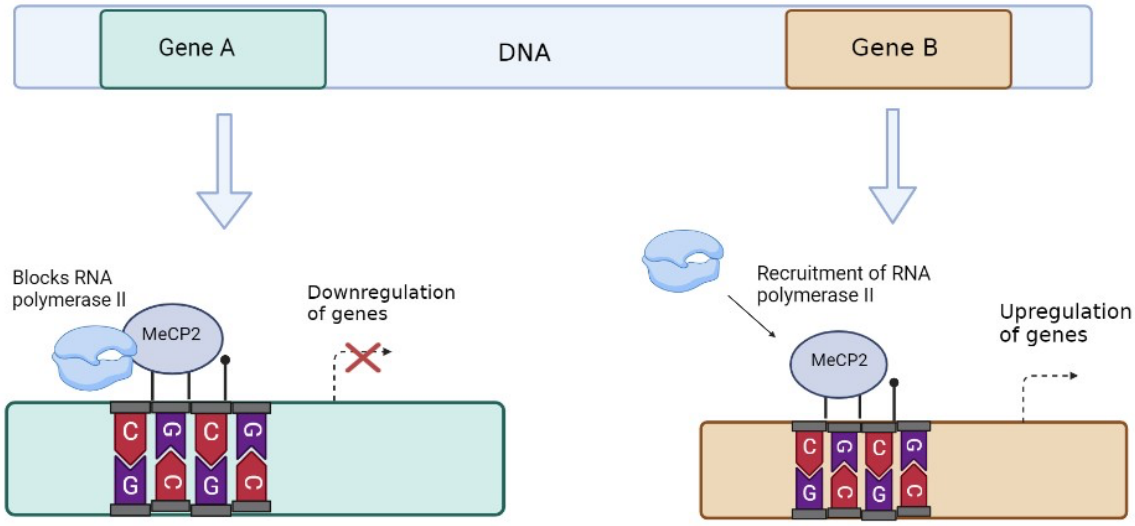


Figure 8. A model illustrating MeCP2 binds to methylated CpG site and blocks RNA polymerase at some genes but promote RNA polymerase activity at other genes.

Methods

Transformation

MeCP2 was expressed from pEX-A128 vector with Ampicillin antibiotic resistance gene (Created using gene synthesis, Genscript). mSSSI was expressed on the plasmid pAIT, a gift from NEB. The catalytically inactive from mSSSItc was created by removing the C-terminal portion of mSSSI from the vector using restriction digest, which abolished all ability to methylate DNA (Sheeba Singh and Peter Sarkies, unpublished data).

The C2523 strain of *E. coli* bacteria were used for transformation. The strain is deficient in proteases Lon and OmpT and resistant to phage T1 and importantly, lacks the MrABC operon, meaning that it does not restrict methylated DNA and so can be used to express CG methyltransferases. To make competent cells, *E. coli* bacteria were grown in Lucia Bertani (LB) medium. Single colonies were inoculated into 2ml of LB overnight. 2.5ml of overnight culture were inoculated into 97.5ml of LB and incubated on a shaker at 37c and 200rpm for ~1.5 hour until the optical density of suspension at 600nm reached in the range of 0.6 to 0.8. Bacteria strains were pelleted by centrifugation of 2 50ml bacteria culture stored in falcon tubes (3750rpm, 5minutes at -4c), removing supernatant and resuspend with 4ml $CaCl_2$, followed by another centrifugation (4000rpm, 1minute at -4c) and resuspension in 50ul $CaCl_2$.

Transformation was by the heat shock method. ~1ug DNA was added to 2.5ml bacterial cell suspension and stored in eppendorf tube on ice for 30 minutes, placed at 42c in heat block for 30 seconds and stored back in ice for 5 minutes. 750ul of LB were then added into each eppendorf tube and incubated on a shaker at 37c and 200rpm. 100ul of bacteria were plated onto appropriate antibiotic selection plates.

Growth Curve Experiments

3 single colonies from each treatment are inoculated into 4-5ml of LB. 0.2mM and 0.5mM IPTG are prepared from 1M IPTG through simple dilution and added into 10ml LB. 2 technical replicates of 200ul of LB with its respective concentration of IPTG is pipetted into a 96-well plate. Only 1 technical replicate of LB without IPTG is used. 200ul of overnight bacteria culture is pipetted into another 96well plate. The bacteria culture is then transferred from the 96-well plate to the other 96-well plate with LB using a replicator. The 96-well plate is then placed into a growth curve machine (Tecan Spark) for growth curve experiment. The growth curve machine is configured to measure the OD600 of bacteria culture every 5 minute where the bacteria is incubated in the machine at an average of 37c and shaken at 216rpm overnight.

The data retrieved were then analysed using one-way ANOVA in GraphPad Prism version 9.2 for Windows, GraphPad Software, San Diego, California, USA, www.graphpad.com. The OD600 of *E. coli* at 8th hour were collected and the growth rate (OD600/h) is calculated by dividing the OD600 by 8. The results were then visualised by plotting boxplots in GraphPad Prism. Kruskal Wallis test is used to see overall differences across several conditions and does not assume normality (non-parametric). Paired t-test following Bonferroni multiple correction test is performed to compare WT and mSSSI and MeCP2 with WT and mSSSI, WT and MeCP2 and WT and mSSSItc and MeCP2. The growth rate fold change of WT and mSSSI and MeCP2, WT and mSSSI and WT and MeCP2 relative to WT at 8th hour is measured by dividing the growth rate of each treatment by the growth rate of WT.

RNA Extraction

3 technical replicates and 3 biological replicates of RNA samples were extracted for each condition using the QIAGEN RNeasy Protect Bacteria Mini Kit (Cat No. / ID: 74124). Only two technical replicates were used from the samples of the 1st biological replicate of WT and mSSSI and MeCP2 and WT and mSSSItc and MeCP2 due to the poor quality of one of the samples. A total of 52 samples were sent for RNA sequencing by the LMS Genomics Facility.

Data Analysis

We used DESeq2 to identify the significantly differentially expressed genes (DEGs) relative to WT in all conditions. We then used the R packages EnhancedVolcano (Blighe, Rana, and Lewis 2018), ComplexHeatMap (Gu et al., 2016) and circlize to create visualisation of the DEGs identified. The packages clusterProfiler (Yu et al., 2012) and AnnotationDbi (Pagès, H., et al. 2021) were used to run Gene Ontology (GO) enrichment analysis of the differentially expressed gene sets. We used seqinr to quantify the number of CG sequences in each gene and then normalized this by the length of the gene to give the number of CGs. The CG proportion and gene length of DEGs and non-DEGs were visualised by drawing boxplots. We then find the relationship between log base 2-fold change (log2FC) and CG proportion of genes by dividing the differentially expressed genes caused by WT and mSSSI and MeCP2 into 5 different bins using the package OneR and investigate the effect of log2FC on CG proportion. The github link attached contained the raw data used in our analysis and the custom script which recorded the code used in R software. (https://github.com/SarkiesLab/meCP2Bacteria)

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